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# Optimization of big data analysis resources supported by XGBoost algorithm: Comprehensive analysis of industry 5.0 and ESG performance

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ABSTRACT

To enable state-owned enterprises in Industry 5.0 to better carry out M&A activities, it is important and necessary to provide early warning of M&A risks, which directly affects the interests of both parties and even affects the effectiveness of state-owned enterprise reform. The author proposes the optimization of big data analysis resources supported by the XGBoost algorithm: a comprehensive analysis of Industry 5.0 and ESG performance. Design a comprehensive evaluation system to measure the M&A risk of state-owned listed companies. Using Python programming language to achieve data crawling and processing. Build an early warning model using the XGBoost algorithm. To further evaluate the effectiveness of the early warning model, comparative experiments were conducted. Using multiple linear regression models to study the significant factors of merger and acquisition risk. The experimental results show that the prediction accuracy based on the XGBoost algorithm is 80 %, which performs the best among all models and has stronger reliability and applicability.

*Conclusion:* The return on investment capital, operating profit margin, and net profit from paid consideration are more important and effective in predicting merger and acquisition risks; The total asset turnover rate, return on investment capital, equity balance, and audit quality are more conducive to suppressing merger and acquisition risks.

## 1. Introduction

With the advent of the digital age, enterprises are facing unprecedented challenges and opportunities in big data. Against the backdrop of Industry 5.0 and ESG (Environmental, Social, and Governance) performance becoming a focus of attention, the XGBoost algorithm, as a powerful tool, provides enterprises with the possibility of achieving resource optimization and sustainable development. Industry 5.0 is a new stage of the Industrial Revolution, with the core concept of integrating artificial intelligence, the Internet, and automation technology into the manufacturing industry to achieve intelligent, flexible, and sustainable production models. The arrival of this stage poses challenges for enterprises to effectively utilize big data resources, optimize production processes, and improve resource utilization efficiency. In this context, the XGBoost algorithm has demonstrated excellent performance as an efficient machine learning algorithm. It not only can handle large-scale datasets, but also has strong generalization ability and robustness, and can discover potential patterns and patterns in the data [1]. In Industry 5.0, the XGBoost algorithm can discover production optimization opportunities, and reduce costs, and waste resources by real-time monitoring and analyzing big data in the production process. Meanwhile, it can also be applied in scenarios such as supply chain management and equipment failure prediction, providing decision support for the implementation of Industry 5.0. In the field of ESG performance evaluation, the XGBoost algorithm can comprehensively analyze the environmental, social, and governance indicators of enterprises and identify key factors related to performance. This helps enterprises to comprehensively understand their situation in sustainable development and provide data support for formulating improvement measures [2]. With the predictive ability of the XGBoost algorithm, enterprises can also predict future ESG performance, providing decision-making reference for managers. A comprehensive analysis of Industry 5.0 and ESG performance will help enterprises achieve more comprehensive and accurate data-driven decision-making, and drive them towards the goals

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of intelligent manufacturing and sustainable development. The application of the XGBoost algorithm can not only improve operational efficiency and reduce risks of enterprises, but also promote their long-term development and social responsibility fulfillment. This article aims to explore the specific application cases of the XGBoost algorithm in the comprehensive analysis of industrial 5.0 and ESG performance and evaluate its effectiveness. Through this study, valuable references will be provided for big data analysis and decision-making in enterprises, promoting the realization of Industry 5.0 and sustainable development. Meanwhile, this article also aims to inspire more scholars and practitioners to explore the potential application of the XGBoost algorithm in big data analysis [3].

This article proposes a big data analysis resource optimization method based on XGBoost algorithm support, which comprehensively analyzes the ESG performance of industrial 5.0 enterprises. Based on considering factors such as economy, society, technology, and environment, an ESG performance evaluation system for Industrial 5.0 enterprises was constructed, and the XGBoost algorithm evaluation model was used to conduct evaluation research on it. Finally, policy recommendations were proposed to help improve ESG performance in industrial 5.0 enterprises. The main structure of the study is as follows:

The economic aspect has the greatest impact: Economic factors have the most significant impact on the ESG performance of Industrial 5.0 enterprises, with a weight of 0.325, highlighting the importance of the operational level of Industrial 5.0 enterprises in ESG performance evaluation.

Technological innovation and environmental protection are equally important: the weights for social, technological, and environmental aspects are 0.284, 0.269, and 0.122, respectively, reflecting the degree to which Industrial 5.0 enterprises attach importance to technological innovation, environmental protection, and social development.

The number of patent authorizations obtained is a key indicator: among the evaluation indicators, the number of patent authorizations obtained has the greatest impact on the ESG performance of Industrial 5.0 enterprises, while the year-on-year decrease in energy consumption per unit output has a relatively small impact, further emphasizing the importance of technological innovation in ESG performance evaluation.

This study provides an important decision-making basis for industrial 5.0 enterprises to formulate development strategies, improve financing capabilities, and prevent operational risks. By comprehensively analyzing various factors such as economy, society, technology, and environment, and using the XGBoost algorithm evaluation model, the ESG performance of Industrial 5.0 enterprises can be more comprehensively understood and evaluated, providing guidance and support for the sustainable development of enterprises.

This article unfolds as follows: Section 2 covers relevant work; Section 3 provides a detailed introduction to our method; Section 4 introduces the implementation details, experimental results, and analysis. Finally, the conclusion is drawn in Section 5.

## 2. Literature review

In recent years, with the rise of Industry 5.0, enterprises have increasingly emphasized environmental protection, social responsibility, and good governance while pursuing economic benefits. Therefore, studying the ESG performance of Industrial 5.0 enterprises has become a focus of attention in both academia and industry. The XGBoost algorithm, as an efficient ensemble learning algorithm, has a wide range of applications in big data analysis. Domenteanu, A. conducts a detailed bibliometric analysis of existing papers belonging to machine learning and big data, pointing out their capabilities from a scientific perspective, explaining the usability of applications, and determining what is practical in constantly changing fields [4]. Li, X. believes that integrating big data analysis into logistics operations has become a transformative approach to improving efficiency, reducing waste, and mitigating environmental impacts [5]. Ahmed, R. used data from ASSET4 and the Global Sanctions Database (GSDB) and found that governments and foreign investors significantly promoted ESG performance, while household investors had a negative impact. In addition, the survey results show that when facing financial sanctions, the reactions of governments, and foreign and household investors vary [6]. Hunting, Z. proposed a hybrid algorithm combining Similar Days (SD) selection, Empirical Mode Decomposition (EMD), and Long Short Term Memory (LSTM) neural networks to construct a Short-Term Load Forecasting Model (SD-EMD-LSTM). Using a weighted k-means algorithm based on extreme value gradient enhancement to evaluate the similarity between predicted days and historical days [7]. Guo, Z. used the XGBoost algorithm to process feature index data, achieve classification and recognition of cognitive data, and apply it to forward perceptual classification [8]. Xie, Y. proposed a physical model based on the XGboost algorithm to solve the problem of predicting cementing quality in oil and gas wells. Through a physical model, a comprehensive analysis was conducted on the nonlinear, time-varying, uncertain, high-dimensional, missing, and imbalanced characteristics of the influencing factors. Finally, numerical examples were used to verify that the established physical model can effectively predict key factors affecting quality, improve process quality, and reduce unit costs [9]. Mo, H. believes that XGBoost is a recently launched machine learning algorithm that has been proven to be very powerful in modeling complex processes in other research fields [10]. Alok, P. proposed a clustering-based extreme gradient enhancement classifier (AXGBoost). The puzzle optimization method updates the probability of data line points to the clustering detector. Using the XGBoost algorithm to predict credit card approval and comparing it with logistic regression to improve accuracy [11]. Wang, T. proposed a supervised algorithm based on Extreme Gradient Boost (XGBoost) for binary and ternary discrimination [12]. Liu, W. proposed a pipeline safety assessment prediction model based on the XGBoost algorithm. Firstly, two models were established based on XGBoost and adjusted to create a relationship between 9 parameters (such as pipeline inner diameter) and NDE/EA results [13]. Wu, J. used the fuzzy c-means clustering method to determine the sensor position and pipeline area. Simulate different leakage scenarios, based on the XGBoost algorithm, and identify the leakage location and degree by monitoring pressure changes at monitoring points. Taking two hydraulic models of water supply networks as examples for simulation prediction, and comparing them with the backpropagation neural network algorithm, the results show that the XGBoost algorithm can not only identify leakage areas but also predict leaks [14]. Salim, K. combines the XGboost algorithm with the genetic algorithm GA-XGboost to find the optimal hyperparameters of the classifier, improve the efficiency of the classifier, and ensure the normal operation of compressor 103 J and water pump system [15]. Tian, J. proposed an Extreme Gradient Enhancement (XgBoost) method based on Feature Importance Ranking (FIR) for fault classification of high-dimensional complex industrial systems. Use the Gini index to rank the importance of features and select features based on their position in the ranking [16]. Dan, W. proposed using the XGBoost algorithm and IoT data to strengthen cooperation between small and medium-sized enterprises in the regional economy, cultivate innovation, and drive economic development. The role of the Internet of Things and machine learning in promoting sustainable regional economic prosperity [17]. Xue, J. analyzed the advantages and disadvantages of the LSTM algorithm and XGBoost algorithm, and integrated the advantages of the two prediction models to obtain a more accurate prediction model, XGBoost LSTM [18].

Artificial intelligence algorithms include machine learning and deep learning. Deep learning is suitable for research with massive amounts of data. The author's research focuses on the industrial 5.0 M&A problem, which has a small amount of data. A risk warning model for mergers and acquisitions of state-owned listed companies was constructed using the XGBoost algorithm. The warning results were compared with classical machine learning models GBDT (Gradient Boosting Decision Tree), RF (Random Forest), and traditional Logistic (Logistic Regression) models in experiments, demonstrating their superiority in accuracy, and use multiple linear regression models to empirically test the influencing factors.

## 3. Research methods

### 3.1. CART regression tree

In equation (1), R represents the sample, j represents the sample features, s represents the eigenvalues, that is, the samples in the j-th feature that are smaller than the eigenvalues are the left subtree, and the samples that are larger than the eigenvalues are the right subtree. The following equation (1):

$$R_1(j,s)=\{x|x^{(j)}\leqslant s\} \text{ and } R_2(j,s)\{x|x^{(j)}>s\} \tag{1}$$

The core task of the CART regression tree model is to partition the sample space into more representative and correlated subspaces by selecting appropriate features and segmentation points. This process requires finding features that can provide the maximum information gain or the minimum mean square deviation reduction, and determining the optimal segmentation point to make the samples in each subspace more similar and minimize the prediction error of the target variable. The objective function generated by the CART regression tree is shown in equation (2):

$$\sum_{x_i\in R_m}(y_i-f(x_i))^2 \tag{2}$$

Where $x_i$ is the representative value of this region. If this method wants to obtain the optimal segmentation feature j and segmentation point, equation (2) can be transformed into solving equation (3), which is the objective function:

$$\min_{d,s}\left[\min_{c1}\sum_{x_i\in R_1(j,s)}(y_i-c_1)^2+\min_{c2}\sum_{x_i\in R_2(j,s)}(y_i-c_2)^2\right] \tag{3}$$

$C_1$ and $C_2$ are the representative values of the leaf nodes after the final division. At this point, as long as all the segmentation points of all features are solved, the optimal segmentation features and points can be found, resulting in a global regression tree.

### 3.2. XGBoost algorithm ideas and principles

(1) XGBoost algorithm concept

XGBoost learns new functions by continuously adding trees and dropping each tree onto its corresponding leaf node based on sample features. The final predicted value can be obtained by adding up the scores on each leaf node. This idea based on the CART regression tree, combined with gradient boosting and regularization techniques, enables XGBoost to effectively handle various types of problems, and has high predictive performance and generalization ability, as shown in equations (4) and (5):

$$\hat{y}=\varphi(x_i)=\sum_{k=1}^{K}f_k(x_i) \tag{4}$$

$$\text{Where } F=\{f(x)=\omega_{q(x)}\}(q:R^m\rightarrow T,\omega\in R^T) \tag{5}$$

The operation process of XGboost is shown in Fig. 1.

(2) The principle of XGBoost algorithm. The objective function of XGBoost is defined as (6):

$$\Gamma^{(t)}=\sum_{i=1}^{n}l(y_i,\hat{y}_i^{(t)})+\sum_{k=1}^{K}\Omega(f_k) \tag{6}$$

$l(y_i,\hat{y}_i^{(t)})$ represents the loss function related to;

Yi represents the actual value of sample i; $\hat{y}_i^{(t)}$ represents the predicted values of the first t decision trees for sample i together; $\Omega(fk)$ represents the model complexity of the third tree.

So when the t tree is generated, the prediction score can be written into the following equation (7):

$$\overset{(t)}{Yi}=\hat{y}_i^{(t-1)}+f_t(x_i) \tag{7}$$

Therefore, the objective function becomes the following equation (8):

$$\Gamma^{(t)}=\sum_{i=1}^{n}l(y_i,\hat{y}_i^{(t-1)}+f_t(x_i))+\sum_{k=1}^{K}\Omega(f_k) \tag{8}$$

$F_t$ ($x_i$) represents the predicted value of the t decision tree for sample i.

According to equation (3), find the $f_t$ that minimizes the objective function. XGBoost approximates it using its Taylor second-order expansion at $f_t$ = 0, and obtains the following equation (9):

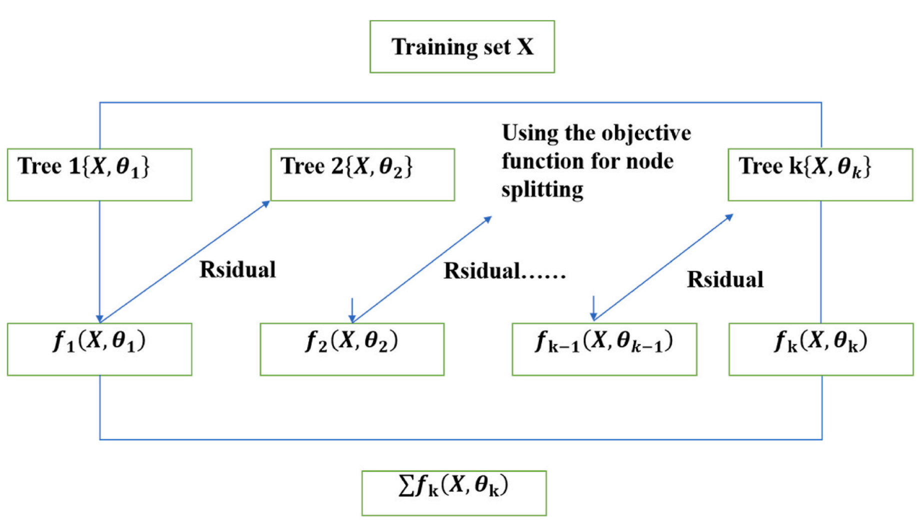



**Fig. 1.** Schematic diagram of XGboost principle.

$$\Gamma^{(t)} \approx \sum_{i=1}^{n} l\left[ {}_{i}, \widehat{y}_i^{(t-1)} + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)\right] + \sum_{k=1}^{K} \Omega(f_k) \tag{9}$$

Among them, $g_i$ is the first derivative, and $h_i$ is the second derivative, as shown in equation (10):

$$g_i = \partial_{\dot{y}(t-1)} l\left(y_i, \widehat{y}_i^{(t-1)}\right), h_i = \partial^2_{\dot{y}(t-1)} l\left(y_t, \widehat{y}_i^{(t-1)}\right) \tag{10}$$

Because the optimization of the objective function is not affected by the residuals and predicted scores of y, the simplified objective function is as follows (11):

$$\widetilde{\Gamma}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)\right] + \sum_{k=1}^{K} \Omega(f_k) \tag{11}$$

As shown in equation (12):

$$\begin{aligned}\widetilde{\Gamma}^{(t)} &\approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)\right] + \\ \sum_{k=1}^{K} \Omega(f_k) &= \sum_{i=1}^{n} \left[g_i \omega_q(x_i) + \frac{1}{2} h_i \omega_q^2(x_i)\right] + \gamma T \\ +\lambda \frac{1}{2} \sum_{j=1}^{T} \omega_j^2 &= \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i\right) \omega_i + \frac{1}{2}\left(\sum_{i \in I_j} h_i + \lambda\right) \omega_j^2\right] + \gamma T\end{aligned} \tag{12}$$

In XGBoost, the objective function is rewritten as a score about leaf nodes ($\omega$). A quadratic function of one variable. This is achieved by introducing the number of control leaf nodes (T) and the score of control leaf nodes ($\lambda$). It is implemented using variables. Specifically, the objective function formula is Equation (13), which uses the vertex formula to find the optimal leaf node score ($\omega$). Meanwhile, equation (14) is used as an evaluation function to compare the gain of the split objective function value with the objective function value of the leaf node. Splitting operation is only performed when the gain exceeds the set threshold.

$$\omega_j^* = -\frac{G_j}{H_j + \lambda} \tag{13}$$

$$\Gamma^{(t)} = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda} + \gamma T \tag{14}$$

In addition, to prevent overfitting, greedy algorithms can be used to determine the structure of the tree and identify the optimal splitting point for each feature. It is also possible to set parameters such as the maximum depth of the tree and sample weights to further control the complexity of the model. XGBoost has multiple parameters, including learning rate, maximum depth of the tree, subsample ratio, etc. These parameters can be adjusted according to the characteristics of the dataset and problem to achieve better performance and generalization ability. There are many parameters in XGboost, as shown in Fig. 2.

### *3.3. Dimension of ESG performance evaluation for industrial 5.0 enterprises*

This article mainly constructs an ESG performance evaluation system for Industrial 5.0 enterprises from four dimensions: economy, society, technology, and environment.

(1) Economic level

The profitability of an enterprise is the fundamental driving force for achieving sustainable development in Industry 5.0. Studying the profitability of an enterprise plays an important role in evaluating the ESG performance of Industry 5.0 enterprises. Promoting the profitability of Industrial 5.0 enterprises not only provides sufficient financial support for their subsequent development but also effectively addresses a series of systemic risks brought about by market uncertainty. In addition, having a good level of profitability also helps to improve the market competitiveness of Industrial 5.0 enterprises. Therefore, this article selects five indicators at the economic level to evaluate the ESG performance of large enterprises, and the specific indicators and meanings are shown in Table 1.

(2) Social level

Society provides a fair and free environment for human survival, therefore, enterprises should fully consider the interests of other stakeholders in society and assume corresponding social responsibilities in their business development process. Unlike other general enterprises, Industrial 5.0 enterprises play an important regulatory role in social development. When social development slows down, Industry 5.0 enterprises can accelerate social development by increasing infrastructure

**Table 1**
Selection of economic indicators for ESG performance evaluation of industrial 5.0 enterprises.

| number | name of index | Metrics choose to mean |
|---|---|---|
| 1 | Basic earnings per share | Measure the effectiveness and profitability of the business operation |
| 2 | net profit growth rate | To measure the potential ability of enterprises to develop their strength and expand their scale |
| 3 | Operating profit margin | Measure the profitability of the business |
| 4 | turnover of total capital | Measure the economic efficiency of enterprise assets |
| 5 | asset-liability ratio | To measure the solvency of enterprises |

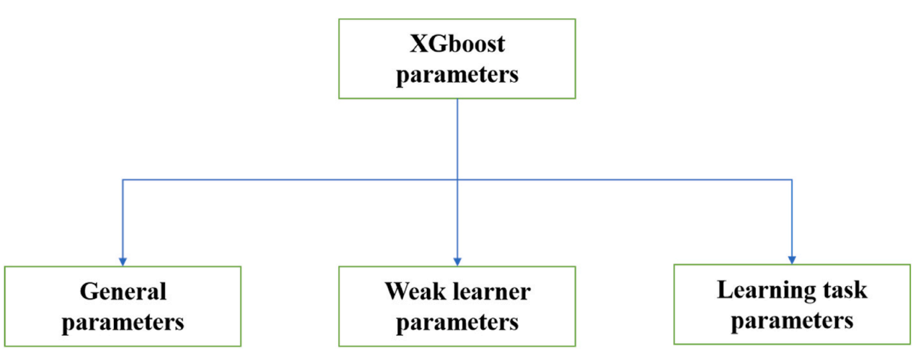



**Fig. 2.** XGboost parameter categories.

construction, creating employment opportunities, and transferring rural labor. Therefore, this article selects four indicators to reflect the ESG performance of Industrial 5.0 enterprises at the social level. The specific indicators and meanings are shown in Table 2.

(3) Technical level

Faced with increasingly fierce market competition in the construction industry, technological innovation has become an effective means for Industry 5.0 enterprises to obtain excess profits. On the one hand, Industrial 5.0 enterprises can effectively enhance their core competitiveness by improving their technological level. On the other hand, Industrial 5.0 enterprises can upgrade their production equipment and improve their construction processes through technological innovation, thereby improving their construction efficiency and reducing their operating costs. Based on this, this article selects three indicators to reflect the ESG performance of Industrial 5.0 enterprises at the technical level. The specific indicators and meanings are shown in Table 3.

(4) Environmental level

The environment can provide the necessary material foundation for human survival and production activities, so Industry 5.0 enterprises should focus on protecting the environment while pursuing benefits. The fulfillment of environmental responsibilities by Industrial 5.0 enterprises reflects their willingness to integrate environmental awareness into their business operations. By cultivating employees to form environmental awareness and establishing corporate environmental concepts, enterprises can help promote sustainable development of the environment. Based on this, this article selects two indicators to reflect the ESG performance at the environmental level of Industry 5.0 enterprises. The specific indicators and meanings are shown in Table 4.

To sum up, this paper specifically selects 14 evaluation indicators from the five dimensions of economy, society, technology, and environment to build the ESG performance evaluation system of industry 5.0 enterprises.

### 3.4. Building an ESG performance evaluation model for industry 5.0 enterprises

This article uses the entropy weight method to calculate the weight of ESG performance evaluation indicators for Industry 5.0 enterprises. The specific steps are as follows:

Assuming there are m evaluation objects and n indicators, construct a matrix of m rows and n columns.

Step 1: Standardization processing of raw data

To prevent errors in the results caused by inconsistent indicator units, the data should be standardized, that is, the negative indicators should be inverted.

**Table 2**
Selection of social indicators for ESG performance evaluation of industrial 5.0 enterprises.

| number | name of index | Metrics choose to mean |
|---|---|---|
| 1 | The growth rate of the number of active employees | To measure the responsibility of enterprises in solving the social employment aspects |
| 2 | Hundred million yuan of output value mortality rate | To measure the responsibility of companies for maintaining social stability |
| 3 | Investment funds in production safety | To measure the responsibility of companies for maintaining social stability |
| 4 | Provide the number of jobs for migrant workers | To measure the responsibility of enterprises in solving the social employment aspects |

**Table 3**
Selection of technical indicators for ESG performance evaluation of industrial 5.0 enterprises.

| number | name of index | Metrics choose to mean |
|---|---|---|
| 1 | research and development expenditure | Measure the intensity of enterprise technological innovation investment |
| 2 | The proportion of professional and technical personnel | Measure the investment ability of enterprises in technological innovation |
| 3 | Number of patents granted | Measure the level of enterprise technological innovation |

**Table 4**
Selection of environmental indicators for ESG performance evaluation of industrial 5.0 enterprises.

| number | name of index | Metrics choose to mean |
|---|---|---|
| 1 | Environmental protection training | Measure the intensity of enterprise environmental protection awareness |
| 2 | Reduction rate of energy consumption per unit of output value | To measure the level of corporate responsibility for energy conservation and environmental protection |

Step 2: Calculate the proportion of index value, as formula (15)

$$P_{ij} = \frac{y_{ij}}{\sum_{i=1}^{m} y_{ij}} \tag{15}$$

Among them, $P_{ij}$ is the proportion of the i evaluated object in the j th evaluation index.

Step 3: Calculate information entropy, as equation (16) (17)

$$k = \frac{1}{\ln m} \tag{16}$$

$$e_j = -k \sum_{i=1}^{m} P_{ij} \ln(P_{ij}) \tag{17}$$

Where, $e_j$ is the entropy value of the j th evaluation index, $0 \leq e_j \leq 1$ , k > 0.

Step 4: Calculate the difference coefficient of the evaluation index, as equation (18)

$$g_j = 1 - e_j \tag{18}$$

Where, $g_j$ is the difference coefficient of the j th index?

Step 5: Calculate the index weight, such as equation (19)

$$w_j = \frac{g_j}{\sum_{j=1}^{n} g_j} \left(0 < w_j < 1\right) \tag{19}$$

Where, $w_j$ is the weight coefficient of the j-th index.

In this paper, the XGBoosts algorithm is used to evaluate the ESG performance of industry 5.0 enterprises.

### 3.5. Selection and analysis of samples and variables

#### 3.5.1. Data selection, processing, and analysis

(1) Merger and acquisition events

Based on the find database of Tonghuashun, the author selected the merger and acquisition events of state-owned listed companies in Shanghai and Shenzhen A-shares from 2014 to 2017 as the research sample. The target of the merger and acquisition is equity, excluding generalized merger and acquisition forms such as asset divestment, asset replacement, asset acquisition, debt revaluation, and absorption merger. After data screening and processing, 1806 merger and acquisition samples were finally obtained. The specific screening and processing steps are as follows: ①Sample of listed companies as acquirers; ② Exclude samples of unfinished or terminated M&A transactions, as well as samples of ongoing M&A transactions; ③ Excluding financial enterprises; ④ Exclude samples with insufficient years of listing and special treatment (ST); ⑤ Exclude samples with incomplete or missing key research data.

(2) Collection of indicator data

There have been numerous normative studies on the risk indicators affecting mergers and acquisitions of state-owned listed companies in the existing literature. However, many governance and supervision indicator data cannot be directly obtained from databases, and obtaining these indicators is difficult and may have a significant impact on the risk warning of mergers and acquisitions of state-owned listed companies. So, the author first selected the iFinD database of Tonghuashun to extract some directly obtainable data, and then based on the principles and methods of Grounded Theory, using Textual Mining and Web Crawler techniques, by listing key indicators, using Python and regularization rules for matching, captured data that listed companies are not easily accessible at first hand. The detailed operation process is shown in Fig. 3. Among them, the application of grounded theory is the process of mining a large amount of empirical data, summarizing and sublimating it into theory; Text mining is a technique that extracts implicit, unknown, and valuable data from massive text databases; Web crawlers are the process of obtaining useful information such as images and data by simulating human access to the network; The regularization rule is a logical formula for string operations, used to match corresponding keywords. In addition, before conducting data capture, it is necessary to create specific rules for indicator capture, namely a requirement document. The requirement document for the three indicator data (COO, FB, WHP) that cannot be directly obtained is shown in Table 5.

(3) Indicator data processing

① Attribute value processing and converting numerical types. Attribute value processing refers to the digitization of nonnumerical attributes. After analysis, it was found that nonnumerical attributes can be divided into two types: Text attributes with trends between attribute values and text attributes with no trends between attribute values. The non-numerical attribute data in the three-level 19 indicator system of key factors for mergers and acquisitions risk of state-owned listed companies all belong to textual attributes with trends between attribute values, such as whether there is a CEO, whether the hired accounting firm is one of the four major accounting firms, and whether there is poverty alleviation behavior. The quantitative methods of 0 and 1 are used to take values. If yes, the value is 1, and vice versa, it is 0. In addition, for some features that are not numerical, these data cannot be directly used by the algorithm, so all nonnumerical data needs to be converted into numerical [19].

② Forward normalization and continuous numerical normalization processing. Before using the XGBoost algorithm to construct a warning model for the risk of mergers and acquisitions of state-owned listed companies, if the indicator direction is not entirely positive or there are reverse indicators, effective conclusions will not be obtained. Therefore, it is necessary to forward process the reverse indicators by changing the reciprocal of the reverse indicators to a positive indicator. The indicators selected by the author, except for the four indicators of accounts receivable ratio, goodwill level, consideration to net profit ratio, and consideration to equity ratio, are all positive, meaning that the larger the indicator value, the better. In addition, due to the large scale of continuous numerical values of some attributes, early warning models have high requirements for the dimensionality of indicators, which greatly affects the convergence speed. Therefore, to make the various indicators comparable and the model's prediction results more accurate, it is necessary to normalize the continuous indicators. The author uses Python software to program according to normalized formulas for processing.

### 3.5.2. Descriptive statistical analysis

(1) Geographical distribution of mergers and acquisitions

In the sample of mergers and acquisitions by state-owned listed companies, the eastern coastal region has the highest number of mergers and acquisitions, while the northeast and central regions have the highest number of mergers and acquisitions; Except for Xinjiang, there are relatively few mergers and acquisitions in the western region. Beijing, Guangdong, and Shanghai ranked among the top three in terms of M&A frequency, with 245, 222, and 201 times respectively, accounting for 13.57 %, 12.29 %, and 11.13 % of the M&A samples, and the total proportion was 36.99 %, exceeding one-third of the total M&A samples. It can be seen that the mergers and acquisitions of state-owned listed companies are highly correlated with local economic development.

(2) Industry distribution of mergers and acquisitions

In the sample of mergers and acquisitions by state-owned listed companies, the manufacturing industry, wholesale and retail industry, and real estate industry are the high-risk areas for mergers and acquisitions, with 796, 176, and 163 times respectively, accounting for 44.08 %, 9.75 %, and 9.03 % of the M&A samples, and the total proportion is 62.86 %, exceeding 60 % of the total sample. The reason is that from the perspective of the entire A-share listed companies, there is a large sample of companies in these industries [20].

(3) Overall description of merger and acquisition risk indicators

Descriptive statistical analysis was conducted on all evaluation indicators of merger and acquisition risks of state-owned listed companies. Table 6 lists the 25 % percentile, median, 75 % percentile, mean, standard deviation, minimum and maximum values. The average ratio of paid consideration to net profit is 2.33, indicating that many state-owned listed companies have paid more for mergers and acquisitions than their net profit for the year; The average current ratio is 1.71, with a median of 1.29, indicating that more than half of the state-owned listed companies have lower current ratios than the average level; The

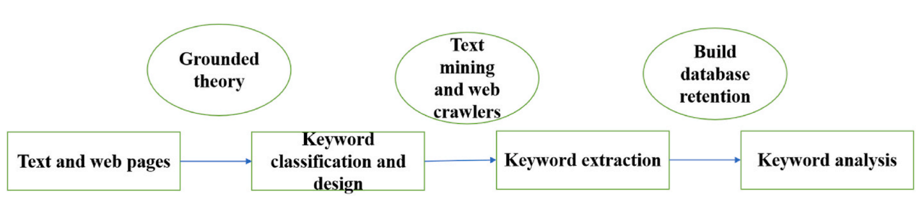



**Fig. 3.** Framework and process for text mining and web crawler data analysis based on grounded theory.

**Table 5**
Requirements document for crawling web pages and text data.

| Matching Rules | Chief Operating Officer (COO) | Audit Quality (FB) | Helping Poverty (WHP) |
|---|---|---|---|
| Enter time range | From 2007 to 2018, if the company goes public later than 2007, the starting time is the year of listing | From 2007 to 2018, if the company goes public later than 2007, the starting time is the year of listing | From 2007 to 2018, if the company goes public later than 2007, the starting time is the year of listing |
| Data source frequency | Annual Frequency | Annual frequency | Annual frequency |
| Data frequency | Annual frequency | Annual frequency | Annual frequency, based on the time of announcement release. Once an announcement is made, it will be captured. If it is not announced in the current year, the results will be the same as the previous year |
| Data source name | Management status | Annual report, annual audit report | Annual report and social responsibility report |
| Data source channels | IFinD of Tonghuashun Company - Shanghai and Shenzhen Deep Data - Shareholder and Management Status - Management Status - Company Senior Management | Tonghuashun Company Announcement Database, searching for the annual "annual report" and "annual audit report" of each listed company | Tonghuashun Company Announcement Database, searching for the annual "annual report" and "corporate social responsibility report" of each listed company |
| Data source type | web page | plain text | plain text |
| keyword | Chief Operating Officer, COO | PwC, Deloitte, KPMG, Ernst&Young | Poverty alleviation, poverty alleviation investment amount, assistance, poverty alleviation, donations |
| Capture Results Format | If the authentication is passed, the value is 1. If it is not passed or cannot be retrieved, the value is 0 | Verified with a value of 1, failed or unreachable with a value of 0 | Verified with a value of 1, failed or unreachable with a value of 0 |

**Table 6**
Descriptive statistics of various variables in M&A risk warning samples.

| explanatory variable | Sample size | descriptive statistics | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | 25 % quantile | median | 75 % quantile | mean value | standard deviation | minimum value | Maximum value |
| Accounts receivable ratio (RS) | 1806 | 0.09 | 0.20 | 0.36 | 0.23 | 0.17 | 0.00 | 0.76 |
| Total Asset Turnover (TAT) | 1806 | 0.31 | 0.53 | 0.86 | 0.70 | 0.71 | 0.02 | 8.72 |
| Goodwill level (GL) | 1806 | 0.00 | 0.00 | 0.02 | 0.04 | 0.09 | 0.00 | 0.78 |
| Current ratio (GR) | 1806 | 0.91 | 1.29 | 1.90 | 1.71 | 1.96 | 0.12 | 33.16 |
| Return on Investment (ROIC) | 1806 | 1.10 | 3.64 | 7.53 | 5.79 | 62.39 | −35.28 | 2636.81 |
| Net profit cash content (NP) | 1806 | 0.00 | 104.74 | 287.33 | 242.39 | 1575.29 | −17047.79 | 16263.16 |
| Operating profit margin (OPM) | 1806 | 0.77 | 5.17 | 13.16 | 5.58 | 37.65 | −1144.32 | 353.65 |
| Interest per Loan (IOBPS) | 1806 | 0.03 | 0.11 | 0.24 | 0.18 | 0.23 | −0.04 | 2.14 |
| Employee compensation ratio (EC) | 1806 | 0.00 | 0.01 | 0.01 | 0.01 | 0.01 | 0 | 0.11 |
| Tax assets (TA) | 1806 | 0.00 | 0.01 | 0.01 | 0.01 | 0.02 | −0.14 | 0.26 |
| Merger and Acquisition Experience (WMAO) | 1806 | 0.00 | 1.00 | 1.00 | 0.62 | 0.49 | 0.00 | 1.00 |
| Income Growth Rate (IGR) | 1806 | −5.08 | 6.49 | 20.13 | 18.70 | 176.18 | −88.25 | 7123.45 |
| Equity balance (EB) | 1806 | 0.21 | 0.45 | 0.96 | 0.68 | 0.67 | 0.02 | 5.04 |
| Chief Operating Officer (COO) | 1806 | 0.00 | 0.00 | 0.00 | 0.21 | 0.40 | 0.00 | 1.00 |
| Audit Quality (FB) | 1806 | 0.00 | 0.00 | 0.00 | 0.10 | 0.29 | 0.00 | 1.00 |
| Helping Poverty (WHP) | 1806 | 0.00 | 1.00 | 1.00 | 0.73 | 0.45 | 0.00 | 1.00 |
| Institutional Investor Shareholders (IIS) | 1806 | 36.91 | 51.38 | 63.97 | 49.47 | 19.76 | 0.00 | 98.49 |
| Payment consideration to net profit ratio (PNPR) | 1806 | 0.02 | 0.19 | 1.02 | 2.33 | 17.86 | −156.65 | 457.98 |
| Payment consideration equity ratio (PC) | 1806 | 0.00 | 0.02 | 0.07 | 0.09 | 0.32 | −3.32 | 5.10 |

standard deviation between net profit cash content and revenue growth rate is relatively large, indicating that state-owned listed companies have significant fluctuations in net profit cash content and revenue growth rate.

### 3.5.3. Correlation analysis

There may be collinear relationships between variables in merger and acquisition risk warnings, which can interfere with the accuracy of the warning. To make the research conclusions more robust, the author also used Kaiser Meyer Olkin (KMO) statistic, Bartlett's sphericity, Pearson correlation matrix, and Variance Inflation Factor (VIF) for correlation validation:

The Sig values of KMO and Bartlett are 0, less than 0.05, but the KMO value is 0.438, less than 0.6, indicating that the variables are not suitable for factor analysis; The maximum VIF value of each variable is 2.37, with a mean of 1.42; The correlation coefficients between Pearson correlation matrix variables are all less than 0.65. Therefore, the above four methods indicate that there is no multicollinearity interference between variables and the selection of variables is reasonable. All tests were obtained through SPSS 20.0 software [21].

## 4. Result analysis

### 4.1. Category aggregation

To strengthen the generalization ability of the training and test sets, the accuracy of the validation classification method is estimated using the K-fold cross-validation method. K-fold cross-validation divides the entire data set into mutually exclusive K subsets, using (K-1) subsets as the training set each time, as shown in equation (15). The author used a five-fold cross-validation method and roughly divided the samples into five parts in a 4:1 ratio, each containing approximately 362 samples [22]. Four out of five samples were selected one by one, resulting in 1444 samples being used as the training set, while the remaining 362 samples were used as the testing set. One part is used to train the classifier, and the other part is used to test the effectiveness of the classifier. Each sample has 19 attributes and 1 label, and a total of 5 training sessions are conducted. The average of the 5 results is calculated to obtain the estimated model accuracy, as shown in Equation (5).

$$\overline{\theta}=\frac{1}{K}\sum_{\mathrm{m}=1}^{K}\theta_{\mathrm{m}}\mathrm{m}=1,2,3,\cdots,K \tag{20}$$

Among them, $\overline{\theta}$ is the average accuracy of the K-fold, $\theta_{\mathrm{m}}$ is the accuracy of the m-th fold, and K is the fold number.

### 4.2. Main parameters and explanation of XGBoost algorithm

The XGBoost algorithm mainly includes three types of parameters, namely universal type parameters, Booster parameters, and target parameters. Among them, the Booster parameter has a significant impact on algorithm performance, and each type of parameter includes multiple specific parameters, totaling up to dozens [23]. The main purpose of these parameters is to improve model performance and prediction accuracy, so parameter configuration and adjustment are also necessary. The main parameters and explanations of the model are shown in Tables 7–9.

### 4.3. Evaluation of ESG performance in industry 5.0 enterprises

By substituting the data into formulas (15) - (19), the weight of the evaluation indicators on the ESG performance of Industrial 5.0 enterprises can be obtained. The economic aspect has the greatest impact on the ESG performance of Industrial 5.0 enterprises, with a weight of 0.325, which reflects the prominent role of the operational level of Industrial 5.0 enterprises in ESG performance evaluation. The weights of the social, technological, and environmental aspects are 0.284, 0.269, and 0.122, respectively, which further reflects the importance of industrial 5.0 enterprises focusing on technological innovation, environmental protection, and social development [24]. The comprehensive evaluation values of ESG performance for Industry 5.0 enterprises from 2017 to 2021 are shown in Table 10. Usually, the closer the comprehensive score index is to 1, the higher the ESG performance of Industrial 5.0 enterprises. Through analysis of Tables 10 and it was found that the ESG performance of Industrial 5.0 enterprises showed a continuous upward trend from 2017 to 2021, reaching its peak in 2021. This indicates that by continuously improving independent innovation capabilities, strengthening social responsibility, and improving business philosophy, the continuous improvement of the ESG performance of enterprises has been ensured, laying a solid foundation for achieving sustainable development in the future.

**Table 7**
General type parameters and explanations for constructing warning models using the XGBoost algorithm.

| Serial Number | Parameter | Parameter values | Parameter Description |
|---|---|---|---|
| 1 | booster | gbtree | Select the base classifier, optional parameters gbtree and gblinear, default to gbtree |
| 2 | silent | 1 | Whether to print model information, 0 represents printing, 1 represents not printing, default is 0 |
| 3 | thread | 4 | Thread count selection, default to maximum available threads |
| 4 | eta | 0.1 | The learning step size, defaults to 0.3, generally ranging from 0.01 to 0.2 |
| 5 | gamma | 0.1 | Minimum loss function value, default to 0 |
| 6 | max_depth | 6 | The maximum depth of the tree, default to 6, is used to control overfitting |

**Table 8**
Booster parameters and explanations for constructing warning models using the XGBoost algorithm.

| Serial Number | Parameter | Parameter values | Parameter Description |
|---|---|---|---|
| 1 | min_child_weight | 3 | The minimum leaf node weight sum, default to 1, is used to control overfitting |
| 2 | lambda | 2 | Weight L2 regularization term parameter, default to 1 |
| 3 | subsample | 0.7 | The proportion of random sampling defaults to 1, user-controlled overfitting |
| 4 | colsample_bytree | 0.7 | Randomly extract feature ratio, default to 1, user controls overfitting |
| 5 | seed | 1000 | Random Seed |

**Table 9**
Target parameters and explanations for constructing warning models using the XGBoost algorithm.

| Serial Number | Parameter | Parameter values | Parameter Description |
|---|---|---|---|
| 1 | objective | multi: softmax | Loss function selection, default to Reg: linear, the author asked XGBoost to use a softmax objective function to handle multi-classification problems |

**Table 10**
Calculation results of comprehensive score index.

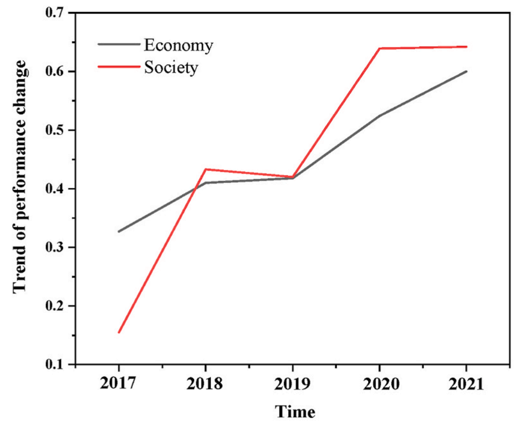
| time | Positive ideal solution distance ( D+ ) | Negative ideal solution distance ( D- ) | Comprehensive score index C |
|---|---|---|---|
| 2017 | 0.691 | 0.168 | 0.196 |
| 2018 | 0.606 | 0.348 | 0.366 |
| 2019 | 0.524 | 0.338 | 0.393 |
| 2020 | 0.330 | 0.498 | 0.602 |
| 2021 | 0.308 | 0.702 | 0.697 |

Table 11 and Fig. 4(a)(b) respectively reflect the ESG performance evaluation results and changing trends of a certain company in the four dimensions of economy, society, technology, and environment from 2017 to 2021.

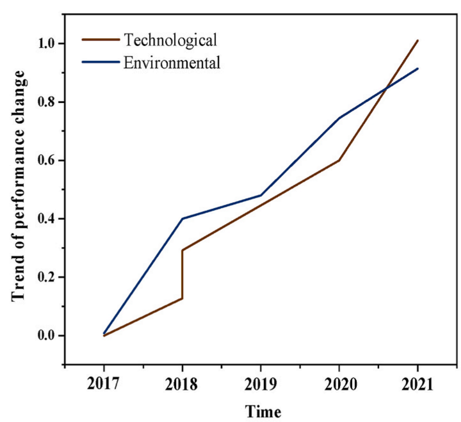
Through analysis, it can be seen that the economic development performance is on the rise from 2017 to 2021, showing a synchronous trend with the overall ESG performance of Industrial 5.0 enterprises. Among them, the operating profit margin has the smallest weight in affecting the economic development performance of Industrial 5.0 enterprises. The overall social responsibility performance of Industrial 5.0 enterprises is showing an upward trend, with the smallest weight being the growth rate of the number of in-service employees. It can be seen that the increase in the number of employees in Industrial 5.0 enterprises has the smallest impact on their social responsibility performance. In addition, the technological innovation performance of Industrial 5.0 enterprises has shown a good trend of increasing year by year in the past

**Table 11**
The ESG performance evaluation results of the economic, social, technical, and environmental dimensions of large construction enterprises.

| time | Economic | social | technological | environment |
|---|---|---|---|---|
| 2017 | 0.324 | 0.149 | 0 | 0 |
| 2018 | 0.414 | 0.435 | 0.130 | 0.399 |
| 2019 | 0.421 | 0.423 | 0.291 | 0.479 |
| 2020 | 0.532 | 0.644 | 0.606 | 0.741 |
| 2021 | 0.603 | 0.654 | 1 | 0.921 |




(a) Change trend of economic and social performance of large construction enterprises




(b) Change trend of technology and environmental performance of large construction enterprises

**Fig. 4.** Trend of economic, social, technological, and environmental performance of large construction enterprises in each year.

five years, with the proportion of professional and technical personnel having the smallest impact on the technological innovation performance of Industrial 5.0 enterprises. At the same time, the environmental protection performance of Industry 5.0 enterprises continues to improve and has a good development trend, indicating that Industry 5.0 enterprises are increasingly emphasizing their responsibility for environmental protection [25].

## 5. Conclusion

The author introduces artificial intelligence algorithms to the field of mergers and acquisitions. Selecting the merger and acquisition events of A-share state-owned listed companies in Shanghai and Shenzhen from 2014 to 2017, this study aims to make assumptions about the influencing factors of merger and acquisition risks for state-owned listed companies, construct a merger and acquisition risk warning model for state-owned listed companies based on the XGBoost algorithm, and explore the importance of each influencing factor on merger and acquisition risk warning. A multiple linear regression model is used to analyze the significance of the relevant factors affecting merger and acquisition risks, the following conclusion has been drawn:

Firstly, the warning results of a state-owned listed company merger and acquisition risk warning model constructed based on the XGBoost algorithm will be compared with classical machine learning models GBDT, RF, and traditional logistic models. The XGBoost model has the best predictive effect on merger and acquisition risk, followed by RF and GBDT, and finally the logistic regression model.

Secondly, the 19 indicators in the constructed indicator system for mergers and acquisitions risk of state-owned listed companies all have an impact on M&A risk warning. It was found that from the overall perspective of the first-level system, asset structure, profitability, external supervision, and M&A characteristic risks are more important for M&A risk warning. Among them, the three indicators of return on investment capital, operating profit margin, and payment consideration to net profit ratio are the most important for M&A risk warning.

Thirdly, the higher the total asset turnover rate and return on investment capital in asset structure risk, the greater the equity balance in corporate governance risk, the four major accounting firms hired in external supervision risk, poverty alleviation behavior, and the higher the shareholding ratio of institutional investors, the more conducive it is to reducing merger and acquisition risk; The higher the interest and tax assets per share in the risk of profitability, and the higher the consideration to net profit ratio and consideration to equity ratio in the risk of M&A characteristics, the greater the M&A risk. At the same time, the indicators of whether it is one of the four major accounting firms and whether it is poverty alleviation were captured using text mining technology.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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